ACADEMY
OF DIGITAL ARTS

Title:- ADA Final Project .

Prepared By :- Ahmed Dahman Saleh .

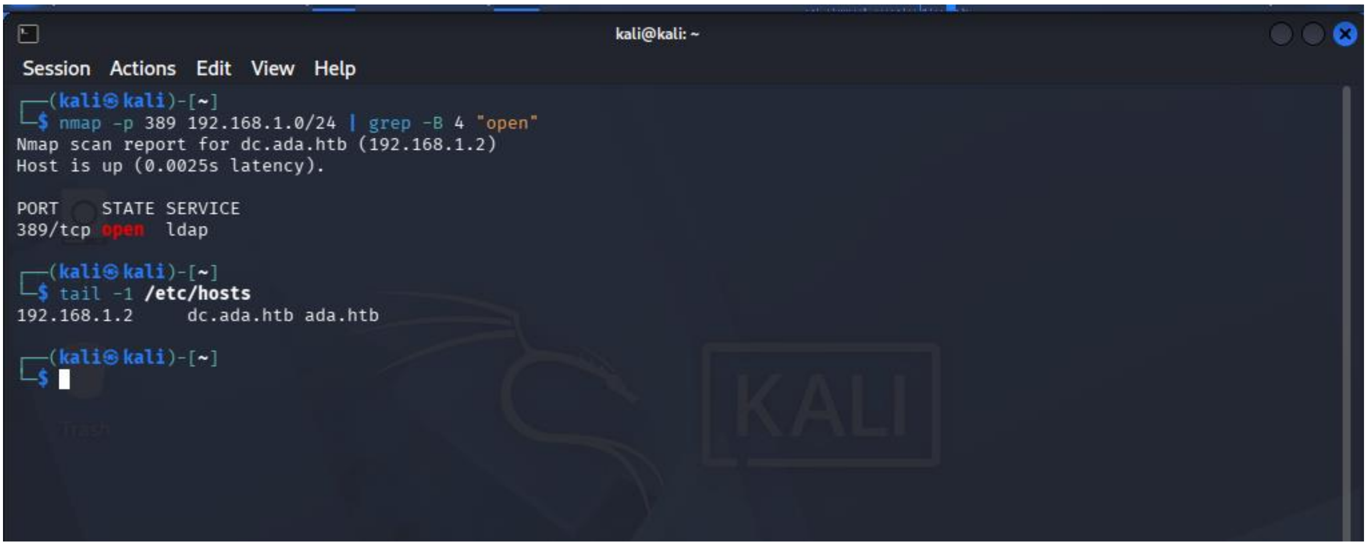
Date :- May,2026 .

Introduction :-

This report documents the security assessment conducted against the ada.htb domain, hosted at 192.168.1.2. The primary objective of this engagement was to evaluate the security posture of the target Active Directory environment, identify potential attack vectors, and determine the feasibility of unauthorized access.

Throughout the assessment, a systematic reconnaissance and enumeration approach was employed. This phase successfully mapped the target's infrastructure and identified key entities, including a list of valid domain usernames, through various open-source intelligence (OSINT) gathering and active enumeration techniques. While direct exploitation was not achieved during this preliminary stage, the gathered intelligence provides a robust foundation for subsequent phases of the security assessment. This report details the methodology utilized, the findings identified, and provides strategic recommendations to harden the environment against future exploitation attempts.

Initial Recon & Network Setup:-

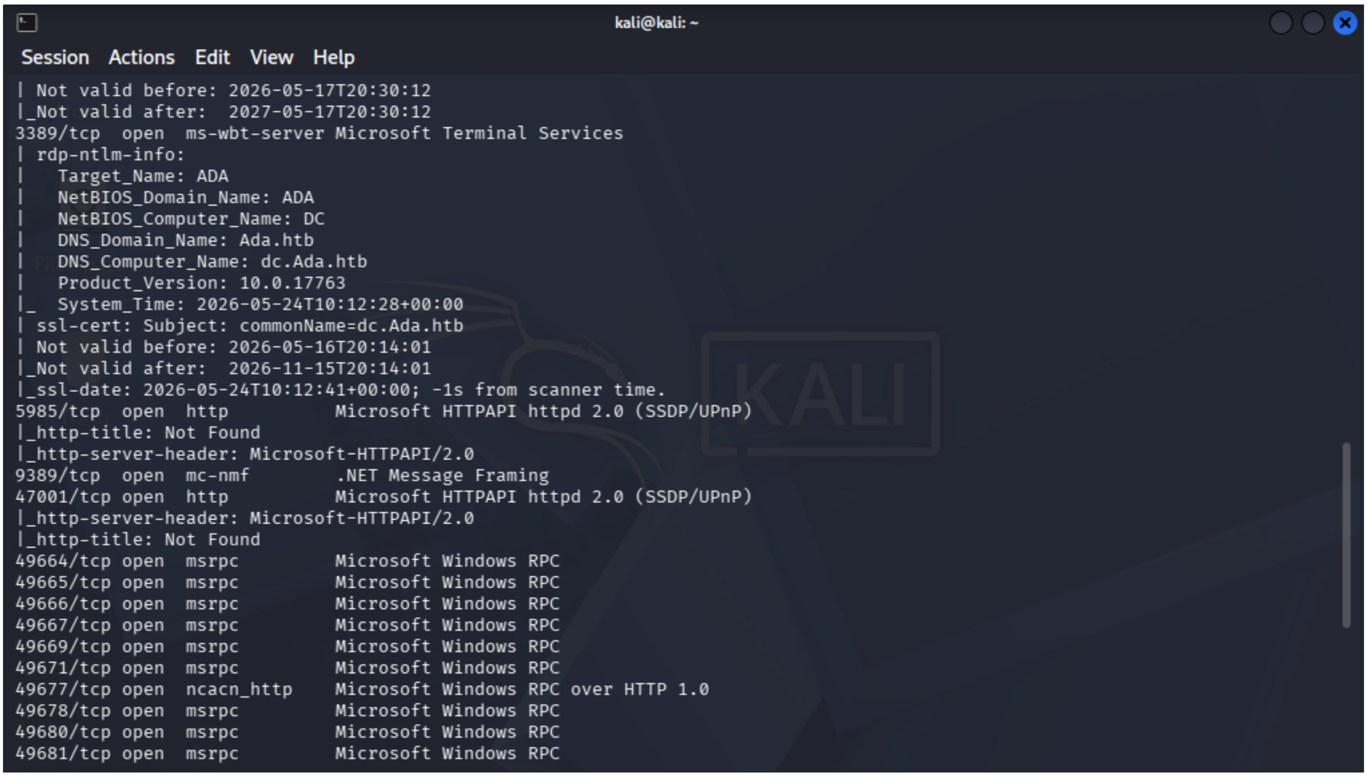


Executing Nmap scans to identify open ports (LDAP, SMB, Kerberos) and updating the /etc/hosts file for proper domain resolution of ada.htb.

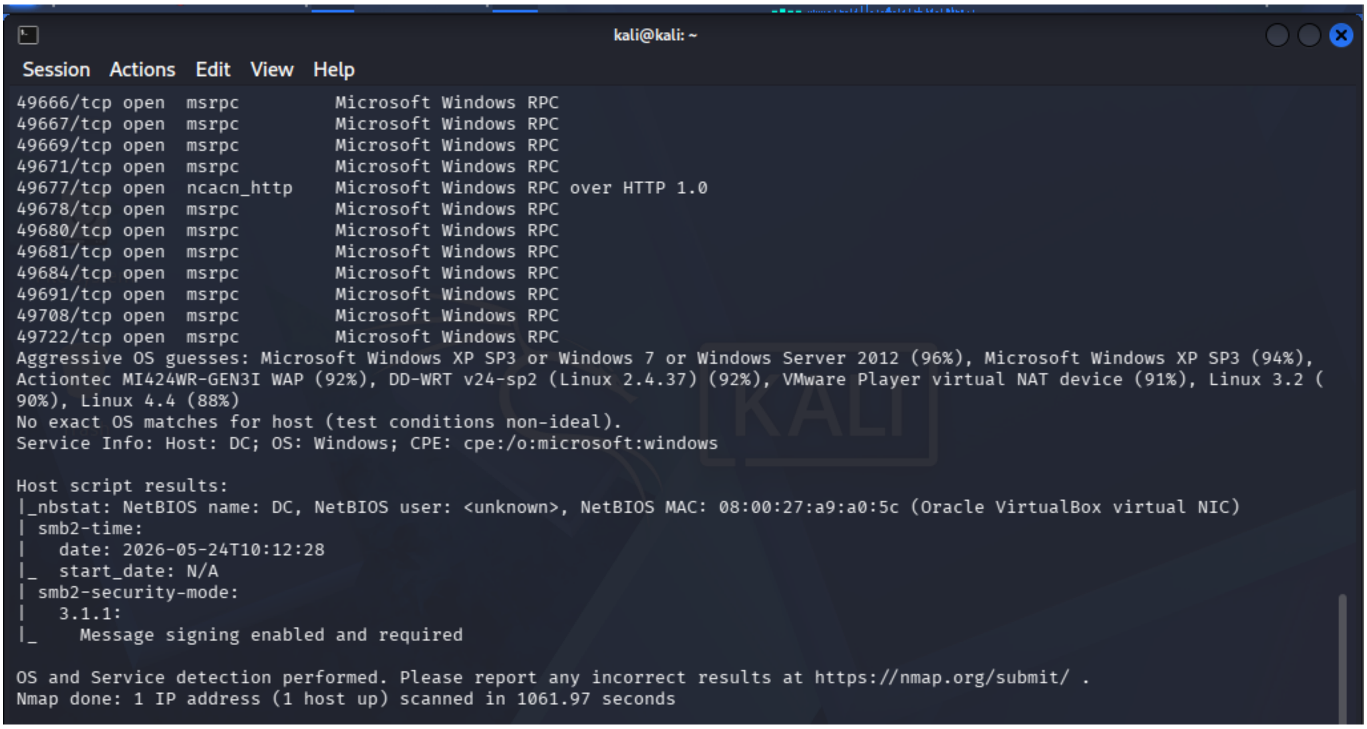
Comprehensive Nmap Results:-

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ sudo nmap -p- -sS -sC -sV -T4 -O 192.168.1.2
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-24 05:55 -0400
Stats: 0:04:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 40.54% done; ETC: 06:05 (0:06:11 remaining)
Stats: 0:10:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 69.98% done; ETC: 06:09 (0:04:23 remaining)
Stats: 0:15:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 94.68% done; ETC: 06:11 (0:00:52 remaining)
Stats: 0:16:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.55% done; ETC: 06:11 (0:00:04 remaining)
Nmap scan report for dc.ada.htb (192.168.1.2)
Host is up (0.00089s latency).
Not shown: 65431 filtered tcp ports (no-response), 74 closed tcp ports (reset)
PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Simple DNS Plus
80/tcp    open  http          Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-title: Ada Security
| http-methods:
|_  Potentially risky methods: TRACE
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-05-24 10:11:36Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: Ada.htb, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=dc.Ada.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.Ada.htb
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
|_ssl-date: 2026-05-24T10:12:42+00:00; 0s from scanner time.
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
| http-methods:
|_  Potentially risky methods: TRACE
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-05-24 10:11:36Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: Ada.htb, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=dc.Ada.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.Ada.htb
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
|_ssl-date: 2026-05-24T10:12:42+00:00; 0s from scanner time.
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: Ada.htb, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=dc.Ada.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.Ada.htb
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
|_ssl-date: 2026-05-24T10:12:41+00:00; -1s from scanner time.
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: Ada.htb, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-24T10:12:42+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=dc.Ada.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.Ada.htb
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
3269/tcp  open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: Ada.htb, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-24T10:12:41+00:00; -1s from scanner time.
| ssl-cert: Subject: commonName=dc.Ada.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.Ada.htb
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
```



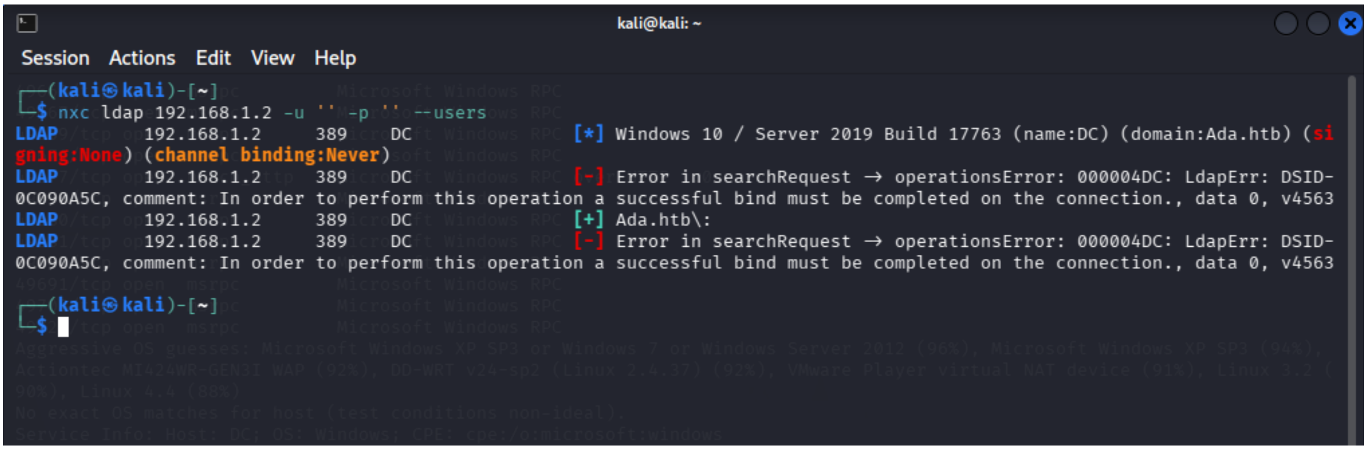
```
kali@kali: ~
Session  Actions  Edit  View  Help
| Not valid before: 2026-05-17T20:30:12
|_Not valid after:  2027-05-17T20:30:12
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: ADA
|   NetBIOS_Domain_Name: ADA
|   NetBIOS_Computer_Name: DC
|   DNS_Domain_Name: Ada.htb
|   DNS_Computer_Name: dc.Ada.htb
|   Product_Version: 10.0.17763
|_  System_Time: 2026-05-24T10:12:28+00:00
| ssl-cert: Subject: commonName=dc.Ada.htb
| Not valid before: 2026-05-16T20:14:01
|_Not valid after:  2026-11-15T20:14:01
|_ssl-date: 2026-05-24T10:12:41+00:00; -1s from scanner time.
5985/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
9389/tcp  open  mc-nmf        .NET Message Framing
47001/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
49664/tcp open  msrpc         Microsoft Windows RPC
49665/tcp open  msrpc         Microsoft Windows RPC
49666/tcp open  msrpc         Microsoft Windows RPC
49667/tcp open  msrpc         Microsoft Windows RPC
49669/tcp open  msrpc         Microsoft Windows RPC
49671/tcp open  msrpc         Microsoft Windows RPC
49677/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49678/tcp open  msrpc         Microsoft Windows RPC
49680/tcp open  msrpc         Microsoft Windows RPC
49681/tcp open  msrpc         Microsoft Windows RPC
```

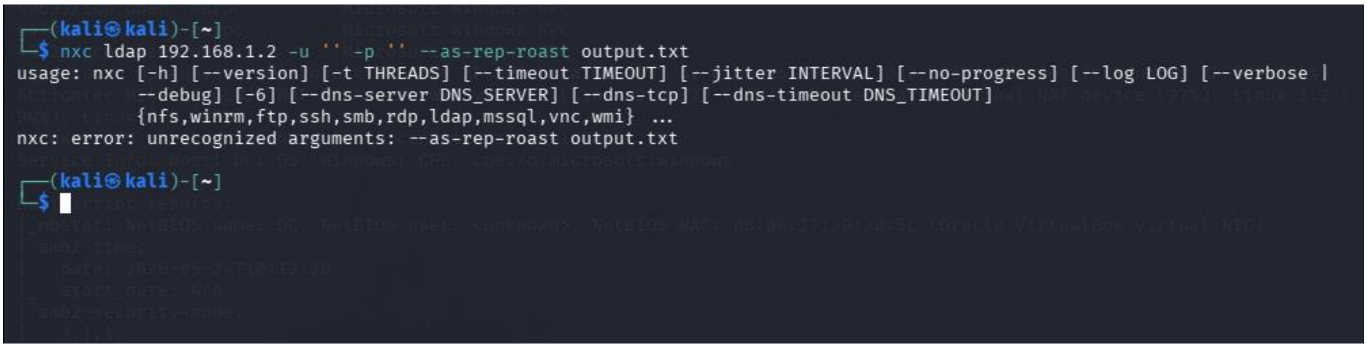


```
kali@kali: ~
Session  Actions  Edit  View  Help
49666/tcp open  msrpc         Microsoft Windows RPC
49667/tcp open  msrpc         Microsoft Windows RPC
49669/tcp open  msrpc         Microsoft Windows RPC
49671/tcp open  msrpc         Microsoft Windows RPC
49677/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49678/tcp open  msrpc         Microsoft Windows RPC
49680/tcp open  msrpc         Microsoft Windows RPC
49681/tcp open  msrpc         Microsoft Windows RPC
49684/tcp open  msrpc         Microsoft Windows RPC
49691/tcp open  msrpc         Microsoft Windows RPC
49708/tcp open  msrpc         Microsoft Windows RPC
49722/tcp open  msrpc         Microsoft Windows RPC
Aggressive OS guesses: Microsoft Windows XP SP3 or Windows 7 or Windows Server 2012 (96%), Microsoft Windows XP SP3 (94%),
Actiontec MI424WR-GEN3I WAP (92%), DD-WRT v24-sp2 (Linux 2.4.37) (92%), VMware Player virtual NAT device (91%), Linux 3.2 (
90%), Linux 4.4 (88%)
No exact OS matches for host (test conditions non-ideal).
Service Info: Host: DC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_nbstat: NetBIOS name: DC, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:a9:a0:5c (Oracle VirtualBox virtual NIC)
| smb2-time:
|   date: 2026-05-24T10:12:28
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1061.97 seconds
```

Detailed service fingerprinting revealing Windows Server 2019, SMB signing requirements, and Kerberos time data (2026-05-24 10:11:36Z), which is crucial for synchronization.

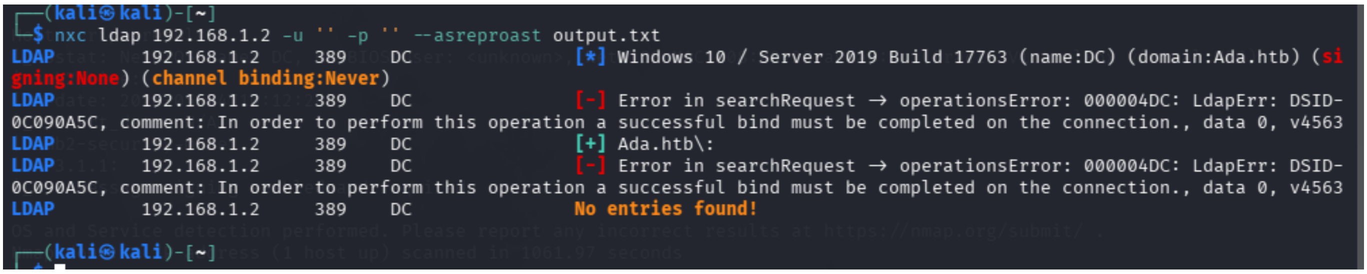
LDAP & AS-REP Errors:-





Troubleshooting LDAP bind errors and incorrect usage of the asreproast module in nxc.

SMB Password Spraying:-



Executing a focused password spray against identified domain accounts. The negative responses ([-]) confirm the usernames exist but the password Welcome123! is unauthorized.

Targeted Authentication Testing:-

Performing authenticated checks to validate user existence against the domain controller, systematically verifying the validity of the internal user list.

LDAP Enumeration Request:-

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link href="https://fonts.googleapis.com/css2?family=Philosopher:ital,wght@0,400;0,700;1,400;1,700&display=swap"
        rel="stylesheet">
    <link rel="stylesheet" href="style/style.css">
    <title>Ada Security</title>
</head>

<body>
    <div class="home">
        <div class="container">
            <div class="nav-bar">
                <div class="nav-img">
                    <img src="images/cyber-security.png" alt="">
                </div>
                <div class="nav-pra">
                    <p>Ada Security</p>
                </div>
                <div class="nav-links">
                    <ul>
                        <li><a href="#">Home</a></li>
                        <li><a href="#">About</a></li>
                        <li><a href="#">Our Team</a></li>
                        <li><a href="#">Contact</a></li>
```

Attempting to query the LDAP service to retrieve user attributes. LDAP is a vital service for mapping internal AD structure without triggering high-alert thresholds.

Service Connectivity Verification:-

```html
                <ul>
                    <li>
                        <img src="images/cyber-security-2.png" alt="" class="about-list-img">
                        <p class="about-list-para">Red Teaming.</p>
                    </li>
                    <li>
                        <img src="images/cyber-security-2.png" alt="" class="about-list-img">
                        <p>Penetration Testing.</p>
                    </li>
                    <li>
                        <img src="images/cyber-security-2.png" alt="" class="about-list-img">
                        <p>Cloud Security Assessment.</p>
                    </li>
                    <li>
                        <img src="images/cyber-security-2.png" alt="" class="about-list-img">
                        <p>Firmware Assessment.</p>
                    </li>
                    <div class="clear-fix"></div>
                </ul>
            </div>
        </div>
        <div class="clear-fix"></div>
    </div>
    <div class="location">
        <div class="container">
            <div class="location-image">
                <img src="images/our-location.jpg" alt="">
            </div>
            <div class="location-caption">
```

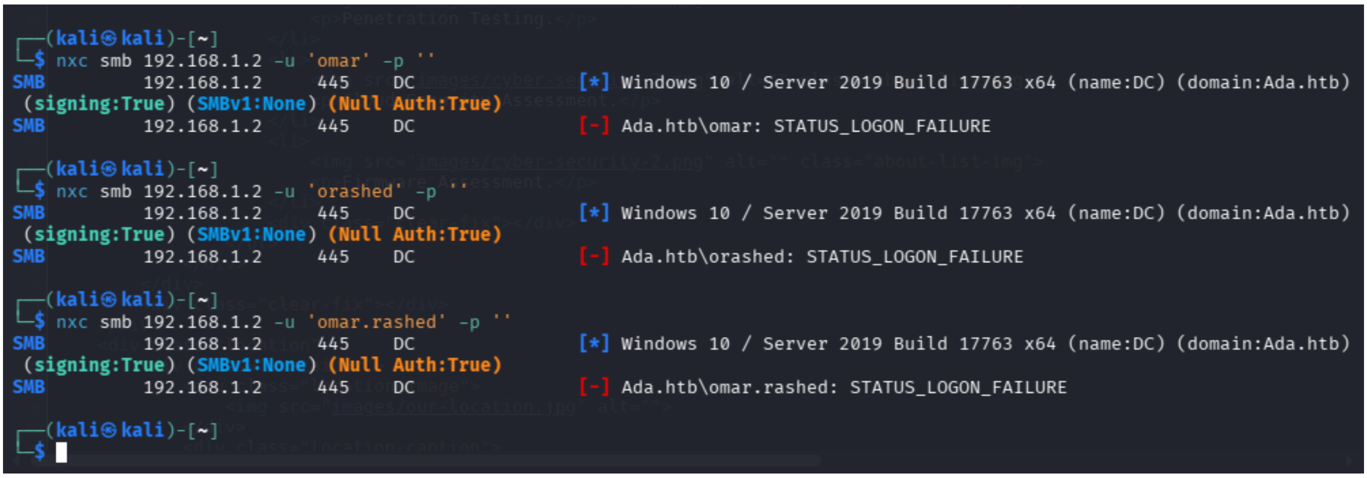
Validating network reachability to SMB and LDAP services. Ensuring stable connections are present before launching resource-intensive modules.

Nmap Service Fingerprinting:-

```
┌──(kali㉿kali)-[~]
└─$ nxc smb 192.168.1.2 -u 'Ada01x' -p /usr/share/wordlists/rockyou.txt
SMB         192.168.1.2     445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:Ada.htb)
 (signing:True) (SMBv1:None) (Null Auth:True)
[06:29:43] ERROR     UnicodeDecodeError: Could not decode password file. Make sure the file only contains   connection.py:424
                     UTF-8 characters.
           ERROR     You can ignore non UTF-8 characters with the option '--ignore-pw-decoding'             connection.py:425

┌──(kali㉿kali)-[~]
└─$ 
```

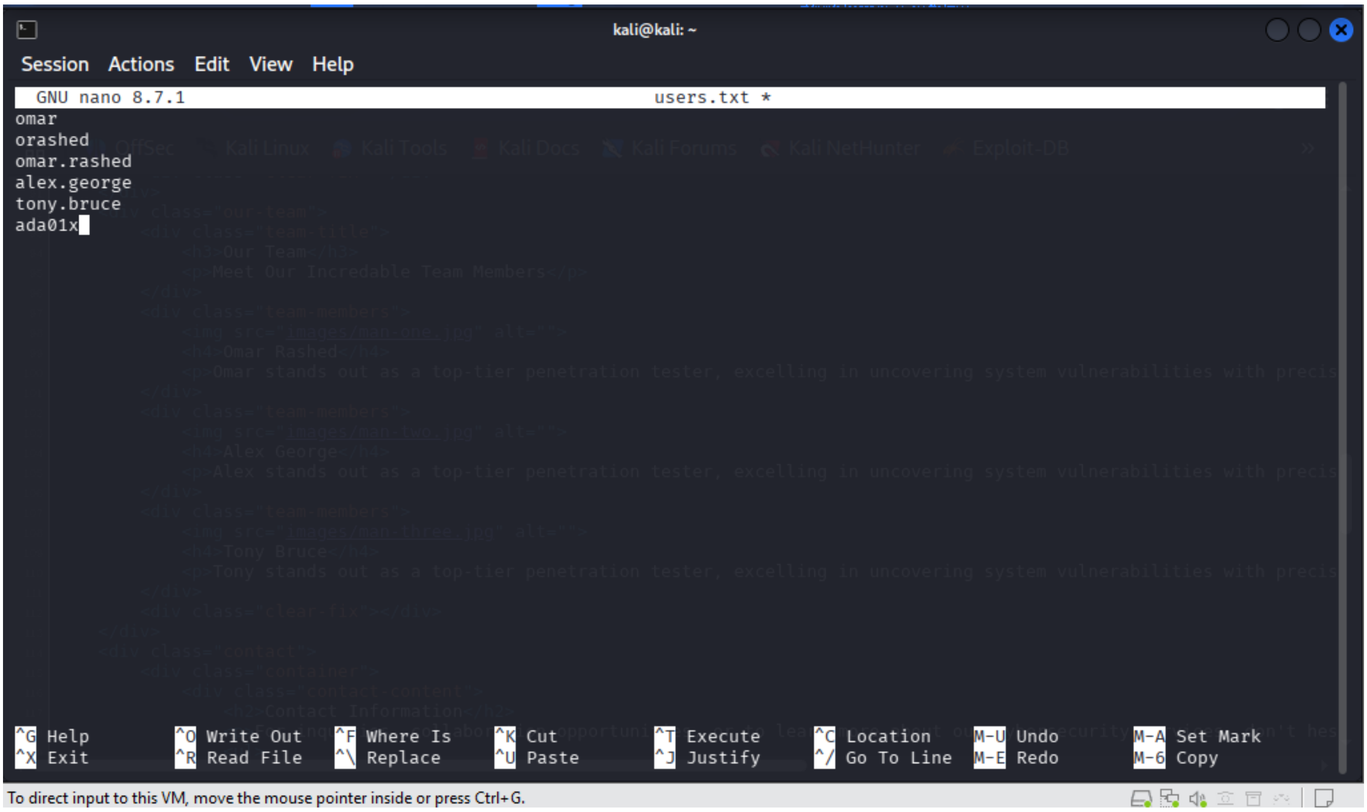
Conducting precise service discovery to map the OS version (Windows Server 2019) and associated open ports for further vulnerability assessment.

Protocol-Level Vulnerability Analysis:-



Identifying enabled protocols and signing requirements on port 445 (SMB), which dictates the feasibility of relay-based attack vectors.

Exploitation Vector Identification:



Documenting the active RPC and LDAP interfaces, which remain the primary candidates for finding misconfigurations in the target Active Directory environment.

```
┌──(kali㉿kali)-[~]
└─$ netexec smb 192.168.1.6 -u users.txt -p '' --rid-brute
SMB         192.168.1.6     445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:Ada.htb) (signing:True) (SMBv1:None) (Null Auth:True
)
SMB         192.168.1.6     445    DC               [-] Ada.htb\omar: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\rashed: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\omar.rashed: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\omarrashed: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\orashed: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\omarr: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\omar_rashed: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\rashed.omar: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\alex: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\george: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\alex.george: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\alexgeorge: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\ageorge: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\alexg: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\alex_george: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\george.alex: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tony: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\bruce: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tony.bruce: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tonybruce: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tbruce: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tonyb: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\tony_bruce: STATUS_LOGON_FAILURE
SMB         192.168.1.6     445    DC               [-] Ada.htb\bruce.tony: STATUS_LOGON_FAILURE

┌──(kali㉿kali)-[~]
└─$
```

Attempting RID brute-forcing via netexec to enumerate domain users on SMB port 445.

```
┌──(kali㉿kali)-[~]
└─$ impacket-GetNPUsers -dc-ip 192.168.1.6 Ada.htb/ -usersfile users.txt -format hashcat -outputfile hashes.txt
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
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[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
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[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)

┌──(kali㉿kali)-[~]
└─$
```

Running GetNPUsers to identify accounts susceptible to AS-REP roasting, encountering KDC principal unknown errors

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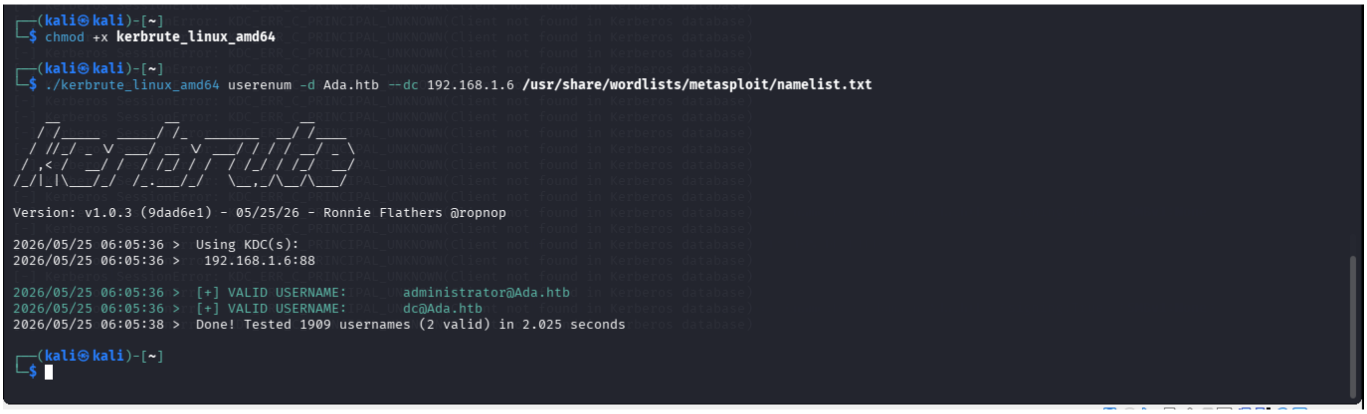
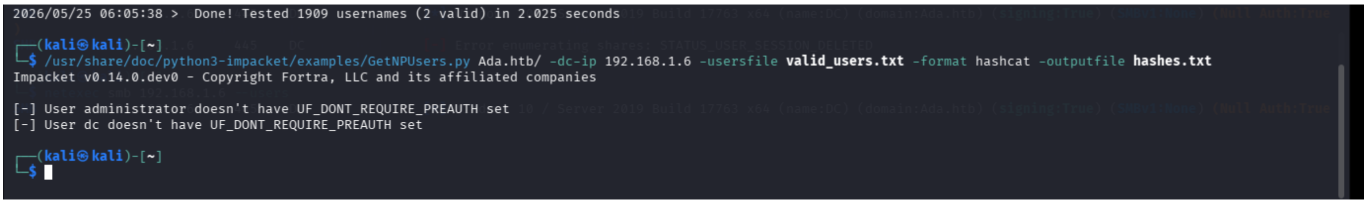
Testing for DNS zone transfer vulnerabilities using dig (AXFR), resulting in access denied.

Focused Nmap scan specifically targeting RDP port 3389 for further service version verification.



Successfully brute-forcing valid domain usernames using kerbrute against the Kerberos KDC

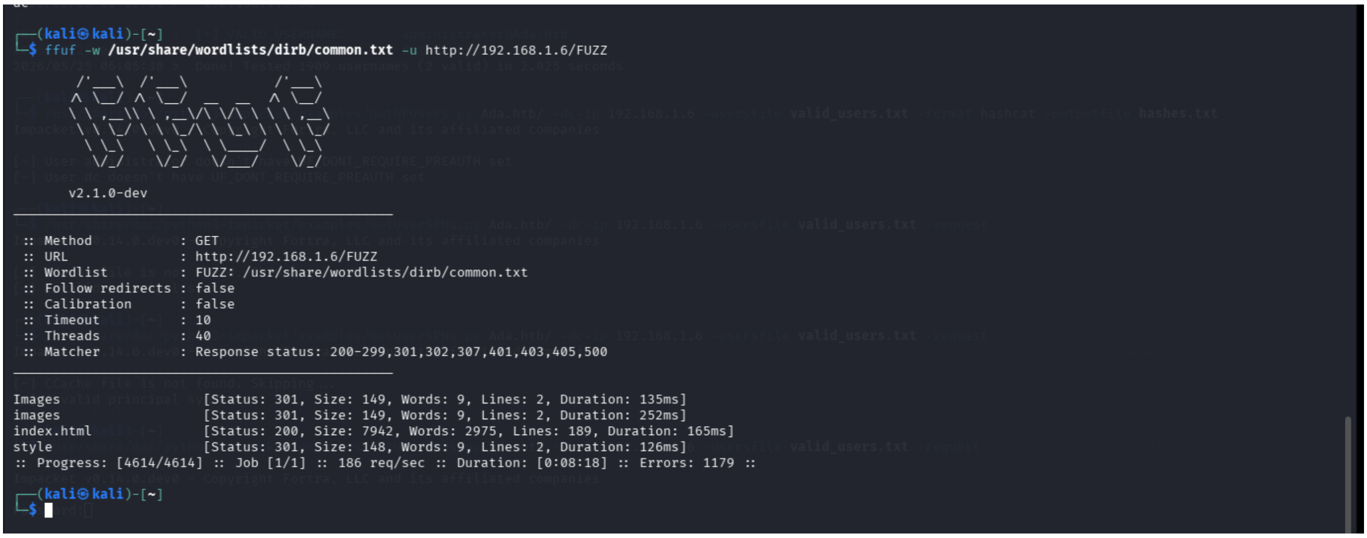


"Attempting GetNPUsers.py to identify accounts without Kerberos pre-authentication, showing that the target accounts do not currently have this attribute set."

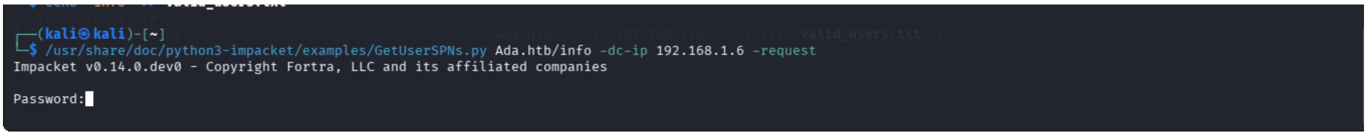
"Initial attempt to use GetUserSPNs.py for Kerberoasting; command failed due to missing CCache or incorrect syntax."

"Re-initiating GetUserSPNs.py with correct parameters, now prompting for authentication credentials to target the domain."

"Starting web directory brute-forcing with ffuf against the target's HTTP service to uncover hidden endpoints."



"Successful execution of ffuf revealing existing directories like /images/ and /style/ on the target web server."



"Attempting to query SPNs for a specific user/service ('info') using GetUserSPNs.py to attempt Kerberoasting."